MIL-A-45877A(AR)
18 JULY 1983
SUPERSEDING
MIL-A-45877(PA)
9 June 1975

MILITARY SPECIFICATION

ADAPTER, CONNECTOR: W5F1 (TRAINING) AND CABLE ASSEMBLY, SPECIAL PURPOSE, ELECTRICAL: W6FE1 (TRAINING)

This specification is approved for use by the US Army Armament Research and Development Center and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers the following components which are used in the Nike Hercules warhead section trainer, Type X:

a. One type of adapter, designated Adapter, Connector: W5F1 (Training). This adapter is fused and is used to prevent damage to warhead section components in the event of missile system malfunctions.

b. One type of cable assembly, designated Cable Assembly, Special Purpose, Electrical: W6FE1 (Trainer). This training cable assembly is used with the ground test equipment and is used to check the condition of circuits in the Type X training warhead section.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Standards. Unless otherwise specified, the following standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS), specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: DRSMC-QAN-Q(D), Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

FSC 1135

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:03Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.
MIL-STD-1460 - Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure for.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (see 6.5)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER

8865392 - Case.
8880573 - Cable Assembly, Special Purpose, Electrical: W6FE1 (Training).
8880574 - Adapter, Connector: W5F1 (Training).
8880615 - Carton, Packing, Ammunition, for Cable Assembly, Special Purpose: W6FE1 (Training).
8880616 - Carton, Packing, Ammunition, for Adapter, Connector: W5F1 (Training).

Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions, should be obtained from the contracting activity or as directed by the contracting officer.

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials and components. Materials shall conform to applicable specifications and drawings. Each component shall conform to all dimensions and tests specified on the component drawing. The contractor shall have available verifiable proof (i.e., objective evidence (see 6.2.1)) that materials and components were fabricated, inspected, and tested under controllable conditions as set forth in the contractor's quality control or inspection procedures.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:03Z
Check the source to verify that this is the current version before use.

3.2 Assembly. The adapters/cable assemblies shall comply with all requirements specified on drawings (dwg.) 8880574 and 8880573, with all requirements specified herein, and with the requirements of applicable specifications to the extent specified on the drawings and herein.

3.3 First article. When specified, sample adapters/cable assemblies shall be subjected to first article inspection (see 4.3 and 6.1.1).

3.4 Continuity and resistance. Each adapter/cable assembly shall exhibit continuity between connected terminals, and the resistance shall be not greater than 0.25 ohm (refer to applicable drawing for wiring diagram). Testing shall be as specified in 4.5.1.

3.5 Insulation resistance. The insulation resistance of each adapter/cable assembly shall be not less than 20 megohms. After humidity testing, the insulation resistance of each adapter/cable assembly shall be not less than 15 megohms. There shall be no evidence of insulation breakdown or other damage during this test. Testing shall be as specified in 4.5.2.

3.6 Dielectric strength. The adapters/cable assemblies shall be subjected to a high-potential alternating current at a frequency of 60 hertz (Hz). The adapters/cable assemblies shall show no indication of dielectric failure as evidenced by flashover (surface discharge), sparkover (air discharge), breakdown (puncture discharge), or inability to attain or maintain the high potential. Testing shall be as specified in 4.5.3.

3.7 High temperature. The adapters/cable assemblies shall withstand a high temperature of 165° ± 5°F, for a minimum period of 4 hours, without evidence of cracking, swelling, distortion, or other damage. After this test (at room temperature), the adapters/cable assemblies shall comply with the requirements of 3.4, 3.5, and 3.6. Testing shall be as specified in 4.5.4.

3.8 Low temperature. The adapters/cable assemblies shall withstand a low temperature of -33° ± 3°F, for a minimum period of 4 hours, without evidence of cracking, swelling, distortion, or other damage. After this test (at room temperature), the adapters/cable assemblies shall comply with the requirements of 3.4, 3.5, and 3.6. Testing shall be as specified in 4.5.5.

3.9 Humidity. The adapters/cable assemblies shall withstand a relative humidity of 95 ± 5 percent and a temperature of 105° ± 5°F without evidence of cracking, swelling, distortion, or other damage. After this test (at room temperature), the adapters/cable assemblies shall comply with the requirements of 3.4, 3.5, and 3.6. Testing shall be as specified in 4.5.6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:03Z
Check the source to verify that this is the current version before use.

3.10 Serialization. A serial number shall be assigned by the contractor to each adapter/cable assembly produced. In no event, shall serial numbers be duplicated, nor shall they give any indication of classified production quantities. Six-letter serial codes (e.g., VIN SAR) shall not be used on these cables.

3.11 Soldering. All soldering of the adapters/cable assemblies shall be in accordance with MIL-STD-1460.

3.12 Workmanship. All parts and accessories shall be constructed, assembled, and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness, marking, cleaning, and freedom of the parts from burrs and sharp edges which may affect performance or result in injury during handling. All parts shall be free of chips, dirt, grease, rust, and other foreign material. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. When the contract requires first article inspection, the contractor shall submit two adapters or two cable assemblies (as applicable), to the testing facility designated in the invitation for bids or request for proposal.

4.3.2 Inspection and testing to be performed. The first article inspection samples shall be inspected for the defects specified in 4.4.2.1 (as applicable), using inspection methods contained therein, and for all the requirements of Section 3. The test requirements of Section 3 are listed in Table I, and the

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:03Z
Check the source to verify that this is the current version before use.

adapters/cable assemblies shall be tested using the test methods specified in 4.5. Inspection and testing shall be performed with the acceptance inspection equipment specified in 4.4.4. The adapters/cable assemblies shall be tested in the order specified in Table I. If any adapter/cable assembly fails to comply with any of the applicable requirements, the first article samples shall be rejected. The Government reserves the right to terminate its inspections upon any failure of the first article samples to comply with the stated requirements. This series of tests is not considered destructive, and samples so tested shall be returned to the lot.

TABLE I. First article tests.

Test	Requirement paragraph
a. Continuity and resistance	3.4
b. Insulation resistance	3.5
c. Dielectric strength	3.6
d. High temperature	3.7
e. Low temperature	3.8
f. Humidity	3.9

4.4 Quality conformance inspection.

4.4.1 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot, and a batch shall mean an inspection batch for the purposes of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor shall be designated or approved by the Government representative.

4.4.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the major defect or the minor defect table as applicable (see Table II). Any defect found in the sample, selected for the major or minor classification, shall reject the lot. Critical defects shall require 100 percent inspection of all items in the inspection lot. If any critical defect is found in an item, the item shall be removed from the lot. The sequence of inspections for the classification of defects need not be followed so long as all the defect inspections are performed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:03Z
Check the source to verify that this is the current version before use.

TABLE II. Inspection lot sampling.

Major defect table		Minor defect table	
Lot size	Sample size	Lot size	Sample size
1-30	all	1-30	all
31-45	29	31-45	27
46-60	35	46-60	30
61-75	38	61-75	33
76-95	42	76-95	35
96-130	46	96-130	37
131-180	50	131-180	39
181-250	52	181-250	41
251-400	54	251-400	42
401 and over	56	401 and over	44

4.4.2.1 Classification of defects. Critical, major, and minor defects are defined in MIL-STD-105.

4.4.2.1.1 Adapter, Connector: W5F1 (Training) (see dwg. 8880574).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
101.	Ability of connector to engage and disengage with mating receptable (2 connectors)	Manual/Gage
102.	Connector improper, improperly attached, or damaged (including missing, loose, or damaged connector terminals (pins or sockets) and damaged insert material) (2 connectors)	Visual/Manual
103.	Any fuse missing, improper, or improperly mounted (3 fuses) (inspect before installing cover)	Visual
104.	Cover improper, improperly attached, or damaged	Visual
105.	Depth of socket or height of pin in connectors (see 6.4)	Visual
106.	Soldering improper (inspect during assembly) (see 3.11)	Visual
Minor:		
201.	Location of polarizing groove in connector with sockets	Gage
202.	Location of polarizing groove in connector with pins	Gage
203.	Casting compound/potting missing or improper (inspect during assembly)	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:03Z
Check the source to verify that this is the current version before use.

204.	Dust cap missing, damaged, or improper (2 places)	Visual
205.	Serialization missing, improper, or illegible (see 3.10)	Visual
206.	Marking missing, improper, or illegible (including pin or socket designations on connector insert material)	Visual
207.	Evidence of poor workmanship (see 3.12)	Visual

4.4.2.1.2 Case (see dwg. 8865392 covering a detail of dwg. 8880573).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
101.	Length, max.	Gage
102.	Height, max.	Gage
103.	Width, max.	Gage
104.	Location of hole in bar (2 bars)	Gage
105.	Diameter of hole in bar, min. (as applicable for each dash number) (2 bars)	Gage
106.	Distance from back side of case to rear of bar, min. (2 bars)	Gage
Minor:		
201.	Brazing missing, improper, or improperly applied	Visual
202.	Chamfer on bars missing or improper (4 chamfers)	Visual
203.	Protective finish inadequate or improper	Visual
204.	Evidence of poor workmanship (see 3.12)	Visual

4.4.2.1.3 Cable Assembly, Special Purpose, Electrical: W6FE1 (Training) (see dwg. 8880573).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
101.	Length of P1 cable section, min.	Gage
102.	Length of P2 cable section, min.	Gage
103.	Ability of connector to engage and disengage with mating receptacle (2 connectors)	Manual/Gage
104.	Any fuse missing, improper, or improperly mounted (21 fuses) (inspect before installing cover)	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:03Z
Check the source to verify that this is the current version before use.

105.	Improper fuse in F12, F13, or F14 location (inspect before installing cover)	Visual
106.	Connector improper or damaged (including missing, loose, or damaged connector terminals (pins or sockets) and damaged insert material) (2 connectors)	Visual
107.	Cover improper, improperly attached, or damaged	Visual
108.	Depth of socket or height of pin in connectors (see 6.4)	Visual
109.	Soldering improper (inspect during assembly) (see 3.11)	Visual
Minor:		
201.	Location of polarizing groove in P1 connector	Gage
202.	Location of polarizing groove in P2 connector	Gage
203.	Cable sleeving cut or damaged (exposing inner conductors)	Visual
204.	Dust cap missing, damaged, or improper (2 places)	Visual
205.	Casting compound/potting missing or improper (including case and connectors) (inspect during assembly)	Visual
206.	Serialization missing, improper, or illegible (see 3.10)	Visual
207.	Marking missing, improper, or illegible (including pin or socket designations on connector insert material)	Visual
208.	Evidence of poor workmanship (see 3.12)	Visual

4.4.2.1.4 Packaging of Adapters/Cable Assemblies (see Section 5).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
101.	Adapter/cable assembly improperly preserved, packaged, packed, and marked (see applicable paragraph in Section 5)	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.12)	Visual

4.4.3 Quality conformance inspection testing. The contractor shall perform all of the tests specified in Table III. The tests shall be performed in the listed sequence. None of these tests

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:03Z
Check the source to verify that this is the current version before use.

is considered destructive, and samples so tested shall be returned to the lot.

TABLE III. Quality conformance tests.

Test	Test classification
a. Continuity and resistance	Major
b. Insulation resistance	Major

4.4.3.1 Continuity and resistance (see 3.4) and insulation resistance (see 3.5). Each adapter/cable assembly shall be subjected to these tests (see applicable paragraph in 4.5). If any adapter/cable assembly fails to comply with the applicable requirements, it shall be classified defective and be removed from the lot.

4.4.4 Acceptance inspection equipment.

4.4.4.1 Major defects and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration procedures designed or specified by the contractor (see 6.1.2). The contractor shall obtain approval of such equipment designs, operating instructions, and calibration procedures prior to use on the contract (see 6.3).

4.4.4.2 Minor defects. The acceptance inspection equipment, operating instructions, and calibration procedures used by the contractor for minor classification inspection shall be approved by the Government representative responsible for acceptance inspection.

4.5 Methods of inspection. The tolerances specified in this specification are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used. Unless otherwise specified, the tests shall be conducted at room temperature (77° ± 10°F). NOTE: During environmental tests, cable connectors shall be protected by mating connectors (e.g., test connectors) or by suitable protective caps.

4.5.1 Continuity and resistance. The continuity and resistance of the adapter/cable assembly shall be measured by use of a Government-approved resistance measuring device which is capable of accurately measuring low resistances in the range specified in 3.4. The adapter/cable assembly shall comply with the requirements of 3.4 during this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:03Z
Check the source to verify that this is the current version before use.

4.5.2 Insulation resistance. The insulation resistance of the adapter/cable assembly shall be measured by using a Government-approved insulation resistance tester at an applied voltage of 500 ± 50 volts direct current. The measurements shall be made between each connector terminal (pin or socket) and all other electrically isolated terminals and connector shells tied together as one terminal (refer to applicable adapter/cable drawings for wiring list/diagram). The test voltage shall be applied instantaneously, and the insulation resistance shall be measured after a 10-second application of the test voltage. The adapter/cable assembly shall comply with the requirements of 3.5 during this test.

4.5.3 Dielectric strength. The adapter/cable assembly shall be subjected to a potential of 500 ± 50 volts rms for a minimum period of 10 seconds. The test potential shall be applied instantaneously between each connector terminal and all other electrically isolated terminals and connectors shells tied together as one terminal (refer to applicable adapter/cable drawing for wiring list/diagram). A Government-approved test setup shall be used, and suitable means shall be provided to indicated the occurrence of dielectric failure in the event that it is not visually evident in the cables. The adapter/cable assembly shall comply with the applicable requirements of 3.6 during this test.

4.5.4 High temperature. The adapter/cable assembly shall be placed into a test chamber, and the temperature of the chamber shall be raised to 165°F. The adapter/cable shall remain at this temperature for a minimum period of 4 hours. After this high-temperature exposure period, the adapter/cable shall be removed from the test chamber, allowed to stabilize to room temperature (without the use of forced cooling), and then tested as specified in 4.5.1, 4.5.2, and 4.5.3. The adapter/cable assembly shall comply with the requirements of 3.7 during this test.

4.5.5 Low temperature. The adapter/cable assembly shall be placed into a test chamber, and the temperature of the chamber shall be lowered to -33°F. The adapter/cable shall remain at this temperature for a minimum period of 4 hours. After this low-temperature exposure period, the adapter/cable shall be removed from the test chamber, allowed to stabilize to room temperature (without the use of forced heating), and then tested as specified in 4.5.1, 4.5.2, and 4.5.3. The adapter/cable asembly shall comply with the requirements of 3.8 during this test.

4.5.6 Humidity. The adapter/cable assembly shall be placed into a dry oven and conditioned at a temperature of 105°F for a minimum period of 24 hours. At the end of this conditioning period, the adapter/cable shall be transferred to a humidity chamber having a temperature of 105°F and a relative humidity of 95 percent. The adapter/ cable shall remain in this chamber for a minimum period of 96 hours. After this exposure period, the

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:03Z
Check the source to verify that this is the current version before use.

adapter/cable shall be removed from the humidity chamber, allowed to stabilize to room temperature (without the use of forced cooling), and then tested as specified in 4.5.1, 4.5.2, and 4.5.3. The adapter/cable assembly shall comply with the requirements of 3.9 during this test.

5. PACKAGING

5.1 Adapters/cable assemblies for interplant shipments (see 6.2.2).

5.1.1 Preservation, packaging, packing, and marking. The adapters/cable assemblies shall be preserved, packaged, packed, and marked in accordance with MIL-STD-1188 or as specified in the contract or purchase order.

5.2 Adapters/cable assemblies for spare (repair) parts.

5.2.1 Preservation, packaging, and marking. Preservation, packaging, and marking shall be in accordance with the applicable drawing listed in Table IV.

TABLE IV

Item	Packaging drawing
W6FE1 cable W5F1 adapter	8880615 8880616

5.2.2 Packing and marking. Packing and marking shall be in accordance with MIL-STD-1188 or as specified in the contract or purchase order.

6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Whether adapter/cable procurement is for spare (repair) parts or for interplant shipments (see 6.2.2).
c. Applicable methods of packaging and packing (see Section 5).
d. First article sample requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:03Z
Check the source to verify that this is the current version before use.

6.1.2 Contract data requirements. Contractor engineering documentation shall be developed as specified by an approved Data Item Description (DID) (DD Form 1664) when specified on a Contract Data Requirements List (DD Form 1423) incorporated into the contract or purchase order. Deliverable data required by this specification is cited in the following paragraph:

Paragraph No.	Data requirement	Applicable DID No.
4.4.4.1	Quality Engineering Acceptance Inspection Equipment Descriptive Documentation.	DID-R-1714 (Tailored)

Copies of data item descriptions, required by the contractor in connection with specific acquisition functions, should be obtained from the procuring activity or as directed by the contracting officer.

6.2 Definitions.

6.2.1 Objective evidence. Records of the contractor's quality control and inspections which can be verified.

6.2.2 Interplant shipments. Shipments of an item from a manufacturer, or from a supply source, to a receiving activity for (a) use of the item in the next higher assembly (during end item production) or (b) for repackaging of the item, in accordance with 5.2, for use as spare (repair) parts.

6.3 Acceptance inspection equipment. The contractor shall obtain approval of equipment designs, operating instructions, and calibration procedures from Commander, US Army Armament Research and Development Center, Attn: DRSMC-QAN-I(D), Dover, New Jersey 07801.

6.4 Inspection of connectors.

6.4.1 Height of pins. In the event a pin does not appear to be the same relative height of adjacent pins, a gage shall be used to determine the variation from normal height. Reject any connector that has a pin that varies by more than 0.060 inch maximum from the normal pin height of the individual connector. The normal height of the individual connector pins shall be determined by calculating the average height from measurements taken of pins adjacent to the pin that appears to be defective. The forward metal edge of the connector shall be used as the reference surface for all measurements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:03Z
Check the source to verify that this is the current version before use.

6.4.2 Depth of sockets. In the event a socket does not appear to be the same relative depth of adjacent sockets, a gage shall be used to determine the variation from normal depth. Reject any connector that has a socket that varies by more than 0.060 inch maximum from the normal socket depth of the individual connector. The normal depth of the individual connector sockets shall be determined by calculating the average depth from measurements taken of sockets adjacent to the socket that appears to be defective. The forward metal edge of the connector shall be used as the reference surface for all measurements.

6.5 Drawings and publications. Drawings and publications listed in Section 2 of this specification under the heading of US Army Armament Research and Development Center (ARDC) may also include drawings and publications prepared by, and identified as, ARRADCOM, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings and publications. Technical data originally prepared by these activities is now under the cognizance of ARDC.

6.6 Changes from previous issue. Asterisks are not used in this revision, to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 1135-A252)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:03Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-45877A	Adapter, Connector: W5F1 (Training) and Cable Assembly, Special Purpose, Electrical: W6FE1 (Training)

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:03Z
Check the source to verify that this is the current version before use.